Science Teacher, 5C

Vasant Valley School, New Delhi

Subject: Science

3139/S5C/0013

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Science Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Science are:

Recollection or recognition of science facts and concepts
Integrating different concepts or information for decision making

Great job nurturing these talents!

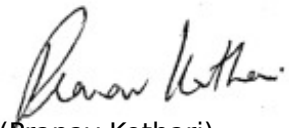
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Science :

Hypothesis formulation or prediction of outcome
Representing, relating or explaining scientific processes or observed phenomena

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student↓	Question No: → / Correct Answers →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Score	Scaled Scores	Percentile	Performance
			D	D	A	A	B	D	C	B	A	C	B	D	D	C	D	A	B	D	A	B	A	B	D	D	B	A	A	B	C	D	A	B	B	D	B				
1	Aarav Yadav		A	C	B	C	C	B	A	✓	C	✓	D	C	A	✓	B	D	A	A	B	C	D	A	B	✓	C	B	✓	✓	✓	✓	B	C	✓	C	C	9	397	13	
2	Aarushi Kumari		B	✓	B	B	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	C	✓	A	D	A	✓	A	C	✓	✓	B	D	A	D	A	✓	✓	✓	✓	✓	20	579	74	
3	Abeer Garg		A	C	✓	✓	D	✓	✓	D	D	A	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	A	B	✓	D	B	✓	✓	✓	A	✓	✓	C	✓	C	21	607	82	
4	Adhvik S Vohra		B	✓	C	B	✓	✓	D	D	B	B	✓	✓	✓	✓	B	B	✓	✓	✓	D	✓	A	B	✓	C	✓	✓	A	✓	✓	✓	✓	✓	✓	A	20	601	80	
5	Akhila S Ratnam		A	A	C	B	✓	✓	✓	A	B	B	✓	✓	✓	✓	B	✓	✓	B	C	D	✓	✓	C	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	A	21	621	85	
6	Akshat Binani		C	✓	✓	D	C	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	C	✓	B	D	C	D	✓	✓	✓	✓	✓	✓	25	664	93	
7	Alysha Nayar		✓	A	✓	B	D	B	✓	A	B	A	D	A	B	A	B	B	✓	A	C	D	✓	A	B	C	D	✓	B	✓	A	✓	D	A	✓	✓	✓	11	428	22	
8	Amaira Chandok		C	C	✓	B	D	✓	A	C	✓	B	A	C	✓	D	B	C	✓	B	✓	D	✓	A	A	✓	C	✓	✓	A	A	B	B	✓	✓	A	C	12	433	23	
9	Amana P Lakhanpal		✓	✓	D	B	✓	C	A	D	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	A	✓	✓	✓	✓	D	B	D	✓	✓	A	✓	✓	✓	✓	A	22	618	84	
10	Anant I Vadehra		✓	C	✓	B	D	C	✓	✓	B	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	C	27	753	99	O
11	Anaya Saikia		B	B	C	B	C	✓	A	✓	D	✓	✓	✓	✓	D	✓	B	A	✓	D	D	B	A	✓	C	D	D	✓	C	✓	✓	B	A	C	✓	✓	14	507	49	
12	Ansh Yadav		✓	B	✓	B	D	C	D	D	D	✓	D	B	✓	✓	✓	C	✓	✓	✓	A	✓	*	✓	✓	A	C	D	✓	✓	C	✓	✓	✓	✓	✓	19	555	66	
13	Anushka Yadav		A	✓	C	C	✓	A	A	✓	C	D	A	B	C	✓	C	B	✓	C	B	C	✓	✓	C	B	C	B	C	C	D	✓	✓	✓	✓	C	D	11	421	20	
14	Aryan Chopra		B	✓	C	D	D	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	D	A	C	D	C	✓	D	C	C	✓	*	✓	C	✓	A	✓	A	16	518	53	
15	Ayanna Baijal		C	✓	✓	B	C	B	✓	✓	C	B	✓	✓	✓	✓	B	✓	✓	✓	✓	C	✓	✓	C	✓	✓	C	B	A	✓	✓	D	C	✓	✓	C	20	580	74	
16	Bhumika		✓	A	C	B	C	A	B	A	✓	B	✓	C	A	✓	✓	C	✓	A	C	✓	✓	✓	C	A	D	C	B	C	B	C	C	A	A	B	D	9	383	10	
17	Himakshi Tiwari		✓	✓	C	B	✓	C	✓	D	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	28	759	99	O
18	Jahnavi Agarwal		✓	A	✓	B	✓	C	✓	✓	D	✓	D	✓	✓	✓	C	C	✓	✓	✓	A	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	C	24	660	93	
19	Juhi Dixit		✓	C	✓	B	D	✓	✓	✓	B	✓	✓	✓	✓	✓	B	B	✓	B	D	✓	C	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	C	✓	24	657	92	
20	Kabir P Kundnani		✓		B	B	D	✓		A	D	✓	D	✓	✓	✓	A	C	D	B	C	✓	✓	C	✓	A	✓	C	✓	✓	A	A	✓	✓	C	✓	✓	16	536	59	
21	Kashvi		✓	✓	B	B	D	✓	✓	✓	D	D	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	A	24	673	94	
22	Krishang G Sekhri		✓	✓	✓	✓	✓	✓	✓	✓	B	D	✓	✓	✓	B	C	C	✓	A	D	A	D	A	B	C	D	B	D	C	A	B	D	C	C	B	C	12	434	24	
23	Maanini Sehgal		✓	C	✓	B	D	C	✓	✓	B	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	D	✓	C	✓	✓	26	726	98	
24	Mandiira Lalie		B	A	*	B	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	A	24	681	95	
25	Neelambar K Sethi		B	C	C	B	✓	✓	A	✓	D	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	A	B	✓	B	✓	C	C	C	21	611	83	
26	Pakhi Lohia		B	✓	D	✓	C	B	✓	D	✓	B	✓	✓	✓	✓	B	B	✓	✓	D	A	✓	✓	C	✓	D	B	D	✓	✓	A	✓	✓	C	✓	C	18	545	63	
27	Pranav Saha		✓	A	C	B	C	A	✓	✓	D	B	✓	✓	✓	✓	B	C	✓	A	D		✓	✓	✓	A	✓	B	D	✓	✓	✓	B	A	✓	✓	✓	18	564	69	
28	Ritika Singh		✓	A	C	B	D	A	✓	✓	D	B	✓	✓	✓	B	A	C	✓	A	C	✓	✓	D	B	A	C	B	D	✓	D	B	D	A	C	A	✓	11	451	29	
29	Sairah Singh		✓	✓	✓	B	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	C	✓	B	✓	✓	✓	✓	✓	A	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	29	744	99	O
30	Tanvie Gogoi		✓	✓	D	D	C	A	*	✓	B	✓	✓	✓	✓	✓	✓	C	✓	B	C	A	✓	A	✓	✓	D	C	C	✓	✓	A	✓	✓	C	B	A	17	535	59	

% answered correct (section)	56	47	38	9	34	47	66	63	19	50	78	78	88	84	50	28	88	44	47	38	84	53	38	69	53	47	56	56	66	66	63	63	50	72	41	579.6/500			
% answered correct (school students)	58	50	39	12	52	47	43	56	27	67	67	77	89	64	39	27	80	40	49	47	71	53	49	61	55	63	47	59	77	69	60	70	57	61	29	573.3/500			
% answered correct (all students)	44	34	35	21	42	51	60	58	21	51	55	65	71	52	35	27	70	47	38	34	57	45	40	56	42	47	42	52	56	47	55	58	42	55	26	508.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**D**	**D**	**A**	**A**	**B**	**D**	**C**	**B**	**A**	**C**	**B**	**D**	**D**	**C**	**D**	**A**	**B**	**D**	**A**	**B**	**A**	**B**	**D**	**D**	**B**	**A**	**A**	**B**	**C**	**D**	**A**	**B**	**B**	**D**	**B**				
31	Yashviraj Solanki		✓	✓	C	B	D	B	✓	✓	D	D	✓	B	✓	✓	A	✓	✓	A	D	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	C	✓	✓	23	632	88	
32	Zorawar S Kalha		✓	✓	B	B	D	✓	✓	✓	D	D	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	A	24	673	94	

% answered correct (section)	56	47	38	9	34	47	66	63	19	50	78	78	88	84	50	28	88	44	47	38	84	53	38	69	53	47	56	56	66	66	63	63	50	72	41	579.6/500			
% answered correct (school students)	58	50	39	12	52	47	43	56	27	67	67	77	89	64	39	27	80	40	49	47	71	53	49	61	55	63	47	59	77	69	60	70	57	61	29	573.3/500			
% answered correct (all students)	44	34	35	21	42	51	60	58	21	51	55	65	71	52	35	27	70	47	38	34	57	45	40	56	42	47	42	52	56	47	55	58	42	55	26	508.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. * Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 5 C, has performed better than the national average in the Science ASSET paper this year.

Class 5	Science	No. of students
5A	113%	34
5B	115%	32
5C (Your Class)	116%	32
All sections combined	115%	98

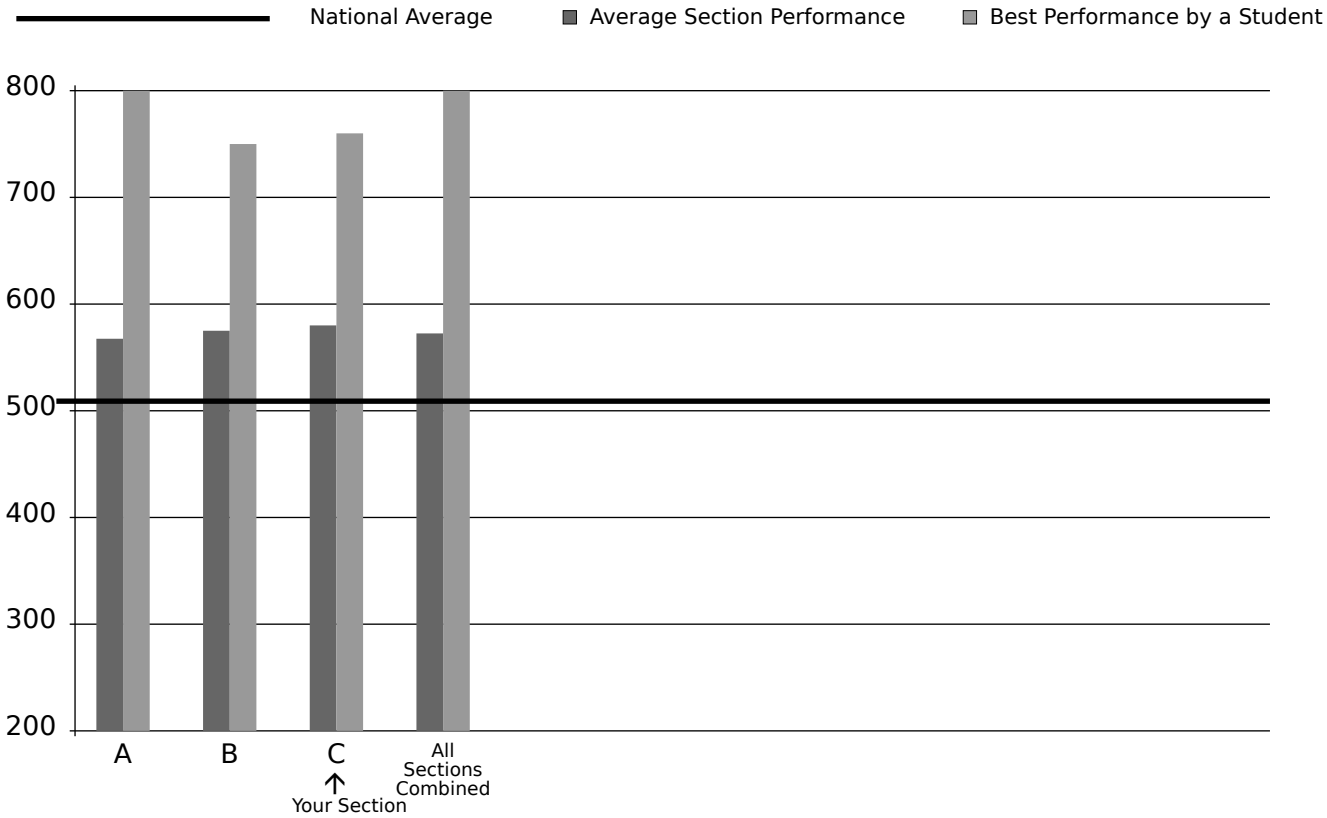
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

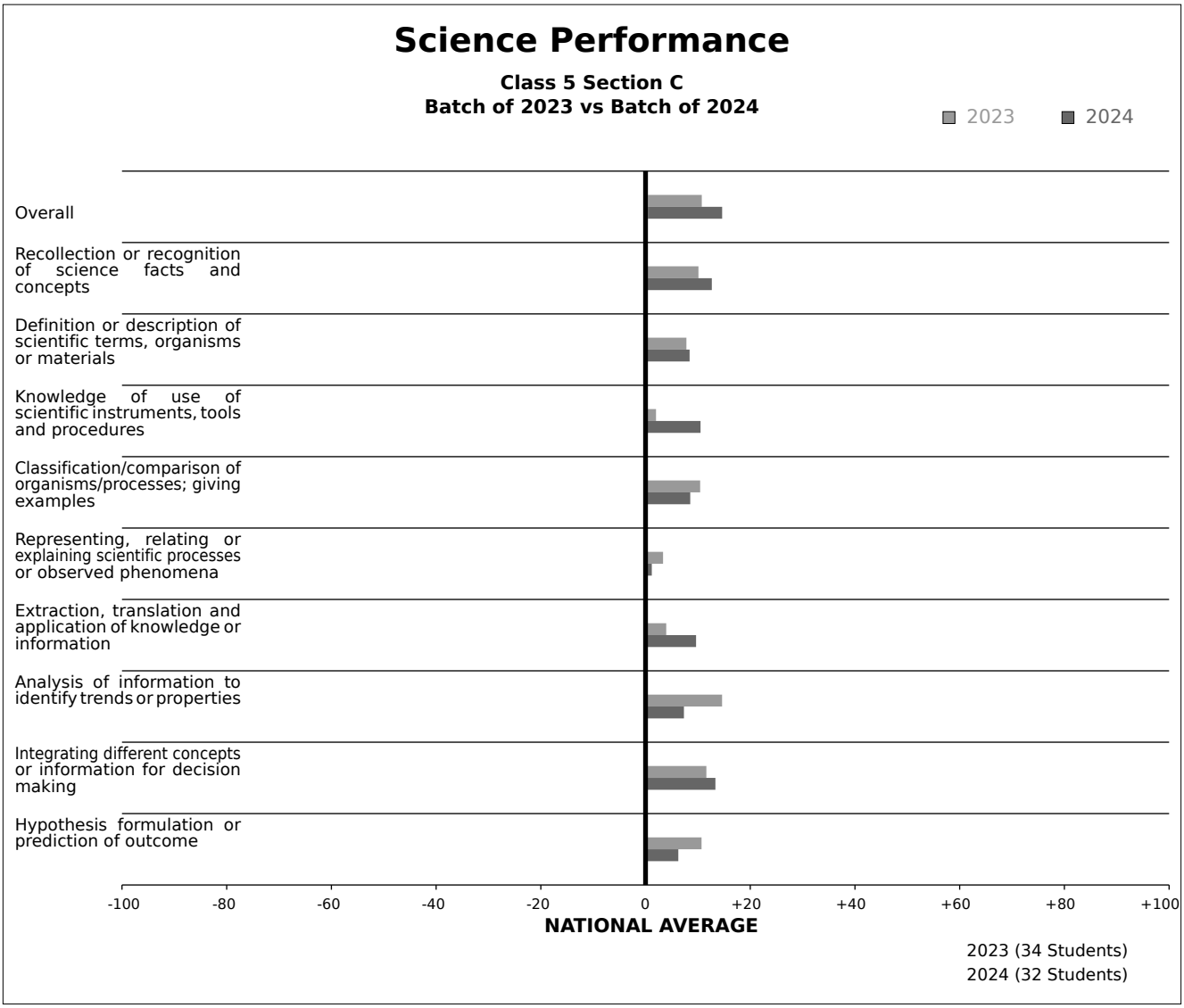
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 5 C Subject: Science Number of Students: 32

S.No.	Skill	Questions	Class 5 C Performance	Class 5 Performance (School)	Class 5 Performance (National)
			Average	Average	Average
1*	Recollection or recognition of science facts and concepts	13,17,26,34	73.4%	73.2%	60.8%
2*	Definition or description of scientific terms, organisms or materials	2,8,22,31	56.2%	54.8%	47.8%
3*	Knowledge of use of scientific instruments, tools and procedures	1,28,35	51.0%	48.6%	40.5%
4*	Classification/comparison of organisms/ processes; giving examples	5,12,16,19,21	54.4%	55.1%	45.8%
5*	Representing, relating or explaining scientific processes or observed phenomena	4,10,29,33	43.8%	53.3%	42.6%
6*	Extraction, translation and application of knowledge or information	7,15,20,27	52.3%	43.9%	42.7%
7*	Analysis of information to identify trends or properties	6,11,18,24	59.4%	53.8%	52.0%
8*	Integrating different concepts or information for decision making	3,14,32	61.5%	57.8%	48.1%
9*	Hypothesis formulation or prediction of outcome	9,23,25,30	43.8%	50.0%	37.5%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Recollection or recognition of science facts and concepts	1.	Hypothesis formulation or prediction of outcome
2.	Integrating different concepts or information for decision making	2.	Representing, relating or explaining scientific processes or observed

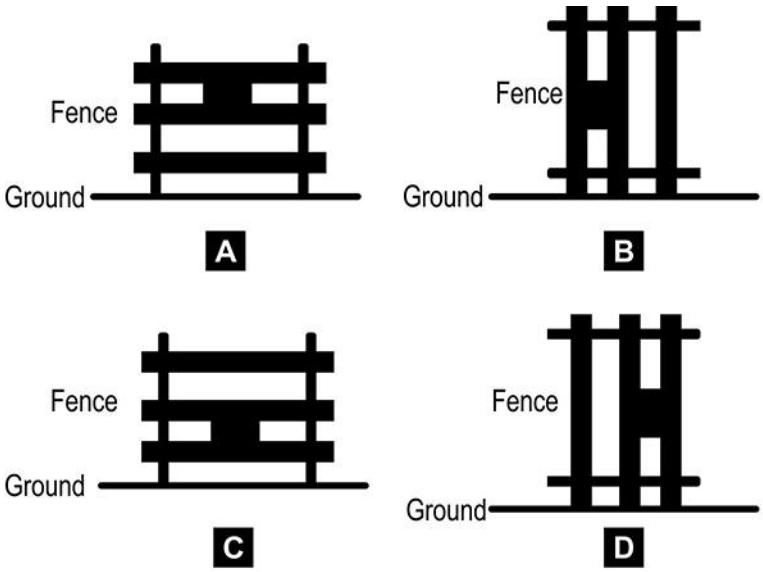
This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 4 The picture below shows the shadow of a fence on the ground.

Shadow of fence on the ground

Which of the following could be the fence?



Skill: Representing, relating or explaining scientific processes or observed phenomena

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
4	35224	A	9%	21%	Your School	
					All Schools	

Q 5 Simon was making a list of things that are man-made. Rama was making a list of things that sink in water.

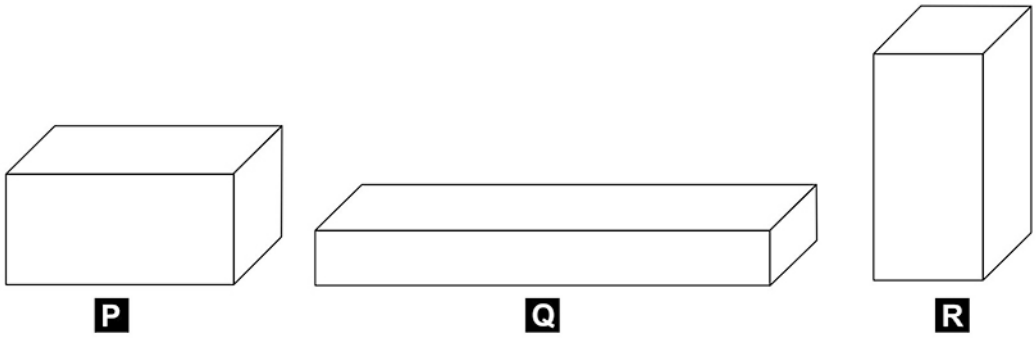
Which of the following can be put in both the lists?

A feather **B** glass rod **C** wooden ball **D** thermocol sheet

Skill: Classification/comparison of organisms/processes; giving examples

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
5	35224	B	34%	42%	Your School	
					All Schools	

Q 6 The three containers shown below have the same capacity. Each of the containers is completely filled to the top one after the other, from a tap with the same flow of water.



(Images not to scale.)

Which of the following is TRUE?

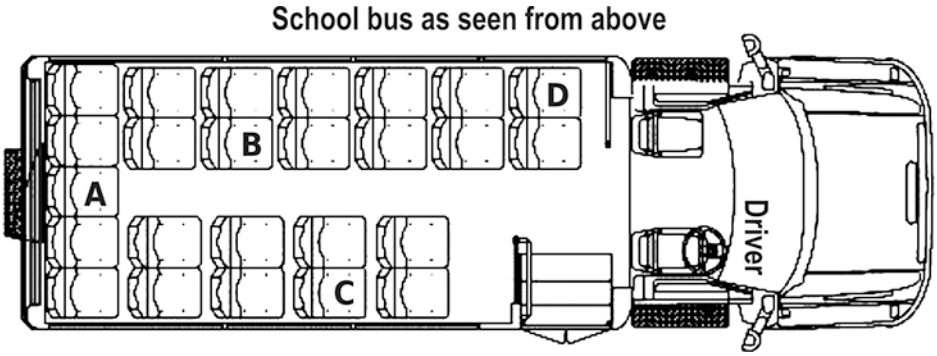
A Container P takes the longest to fill.
B Container Q takes the longest to fill.
C Container R takes the longest to fill.
D All three containers take the same time to fill.

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
6	35224	D	47%	51%	Your School	
					All Schools	

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 14 Rumi goes to school in a school bus every morning. The bus goes directly from her home to the school using Hill Road. Rumi likes the early morning Sun shining on her face on the way to school. The seating arrangement in the bus is as shown here.



On which seat is Rumi likely to get the most sunlight on her way to school?

Skill: Integrating different concepts or information for decision making

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
14	35224	C	84%	52%	Your School	
					All Schools	

Q 21 - I have stripes on my body that help me hide myself from my prey.
- I have soft pads under my paws that help me walk quietly and follow my prey.
- My hind limbs are longer than the fore limbs and help me to jump effectively.
- The length of my body varies from 1.5 - 3 metres.

Who am I?

A tiger **B** snake **C** zebra **D** squirrel

Skill: Classification/comparison of organisms/processes; giving examples

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
21	35224	A	84%	57%	Your School	
					All Schools	

Q 11 The Jacana or Lily Trotter is a bird that can walk on floating lily leaves without sinking! They can do this due to their thin, weak toes which spread their body weight over a large area.

Which of the following feet belongs to the Jacana?

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
11	35224	B	78%	55%	Your School	
					All Schools	

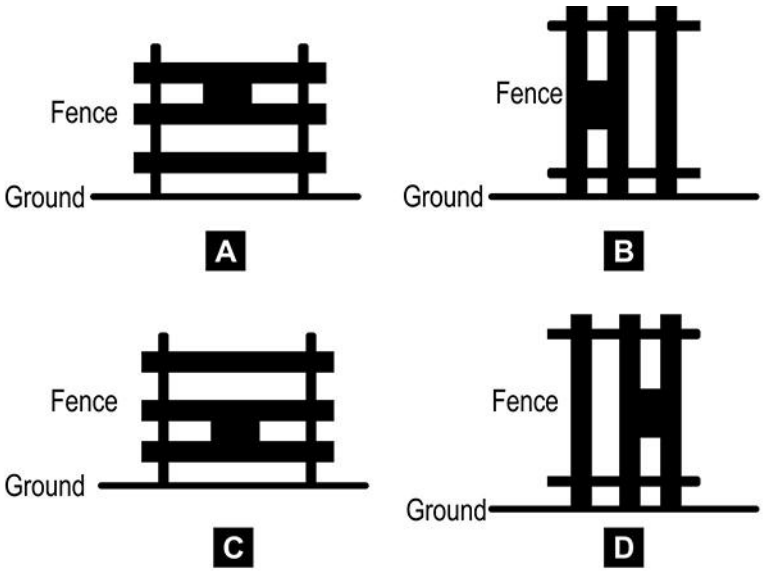
From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 4 The picture below shows the shadow of a fence on the ground.

Shadow of fence on the ground

Which of the following could be the fence?



The question tests students' ability TO relate an object WITH the shadow that it forms. Students choosing the correct OPTION A understand that the light falling ON the fence will CAST a shadow IN the same orientation ON the ground. Students choosing OPTION B might be influenced BY the similar orientation of the shadow AND the fence AND have NOT considered the orientation of the ground. Students choosing OPTION C might be confused about the orientation of the shadow WITH respect TO the object. Students choosing OPTION D might be thinking that the shadow will be reversed AND have also NOT considered the orientation of the ground.

Papercode: 35224, Question no: 4					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	9%	75%	6%	9%	32
National	21%	61%	6%	11%	13500

Q 3 Water is filled to the same level in the jars X and Y as shown in Figure 1. Pebble P is dropped into Jar X and Pebble Q is dropped into Jar Y. The water rises to the same level in the two jars as shown in Figure 2.

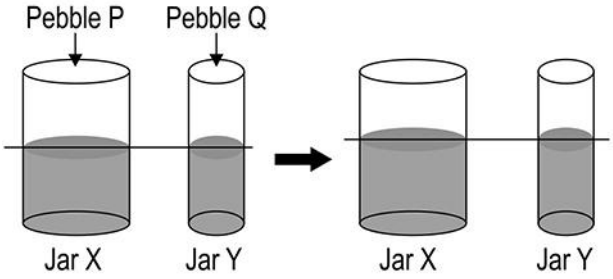


Fig. 1 **Fig. 2**

How does the size of the pebbles P and Q compare?

A Pebble P is bigger than Pebble Q.
B Pebble Q is bigger than Pebble P.
C The size of Pebble P is equal to the size of Pebble Q.
D (Cannot say as jars X and Y are not of the same size.)

The question tests students' understanding of relative volumes. Students choosing the correct option A understand that since Jar X is wider, more water needs to be displaced for the level of water to be raised to the same level as in Jar Y. This is possible only if pebble P is bigger than pebble Q. Students choosing option B might be thinking that the amount of water in Jar Y is lesser, so more water needs to be displaced and hence Pebble Q is bigger. Students choosing option C might be thinking that the sizes of the pebbles are the same as the rise in water levels are the same. Students choosing option D might be thinking that the sizes of the pebbles can be compared only when all the other factors remain the same.

Papercode: 35224, Question no: 3					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	38%	16%	34%	9%	32
National	35%	11%	34%	20%	13500

Q 33 Mansi looked at the Moon on a clear, cloudless night. Shown below is what she observed.

She looked at the Moon again after 2 days.

What could the Moon have looked like?

A

B

C

D

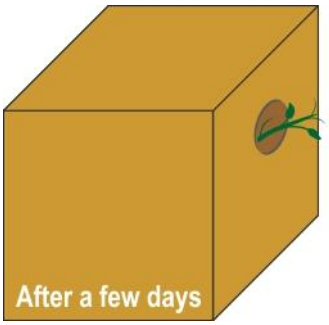
This question was asked to test if students have observed the different phases of the Moon and how they change over the period of a month. Many students have selected the incorrect option C, which indicates that they have probably not observed the phases of the Moon correctly and so are not able to estimate the extent of change of phase in 2 days. It is important that students start making such real life observations which can further be used to get into why those observations hold true. Ask students to draw how the Moon looks at night, every night for a month. At the end of the month, let them describe and discuss what they have observed. Frame some simple questions to check if they have retained what they have observed about the phases of the Moon - like how many days does it take for the new Moon to change to a full Moon, which side of the Moon changes and does the opposite side changes during that same month, etc.

Papercode: 35224, Question no: 33					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	16%	50%	34%	0%	32
National	13%	42%	33%	9%	13500

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Hypothesis formulation or prediction of outcome (44%)

Q 1 Ravi grew a plant in a cardboard box with a single hole on the side. After a few days, the plant started bending towards the hole. Why do you think this happened?

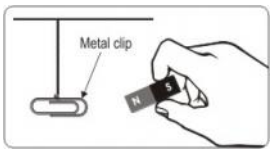


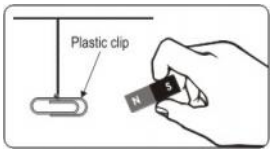
A There is more air near the hole, and the plant is bending towards the air.
B There is more light near the hole and the plant is bending towards the light.
C There is more heat near the hole and the plant is bending towards the hotter region.
D There is more space outside the box and the plant is bending towards the hole to get out.

Explanation:

B: Plants need light in order to produce their food, and therefore grow towards a source of light if they are grown in darkness. They do this so that more light is available to its leaves. The main reason for the plant to move towards the hole will be light and not space. Hence, option D is incorrect.

Q 2 **Akanksha brings a magnet near two clips - one made of metal and another of plastic as shown in the figures.**





Which of these questions is Akanksha probably trying to answer?

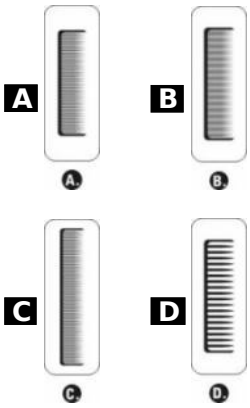
A Do different materials react differently to a magnet?
B Do clips made of plastic and metal attract each other?
C Why does a magnet attract a metal clip?
D Does the size of a magnet affect its power of attraction?

Explanation:

A: Akanksha brings a magnet close to two different things - a plastic and a metal clip - at the same angle and position. This way she can compare how they react to a magnet. Here's why the other options are incorrect: Option B - the plastic and metal clip are never brought close to each other. Option C - you cannot say WHY the magnet attracts the metal clip. Option D - the two magnets are of the same size.

Q 3

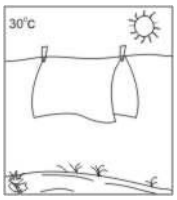
Sparky was combing his hair using the comb shown above. But the knots in his hair were making it difficult for him to comb his hair. Which of these combs will make it easier for him to comb his hair?



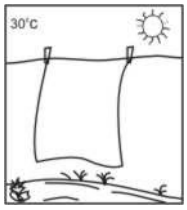
Explanation:

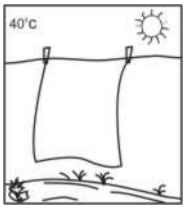
D: Have you tried combing your hair with different types of combs? If you have slightly longer hair, you will know that even if you have very few knots a finer comb, like the ones in options A, B and C will get caught in the knots and it will be difficult to comb your hair easily.

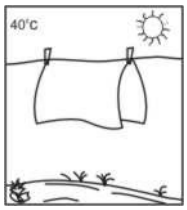
Q 4 Vijay wants to find out whether clothes dry faster on a HOT day. He takes a piece of cloth, folds it (as shown on the right) and notes down the time it takes to dry.

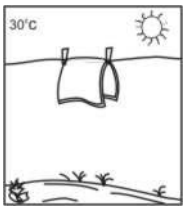


Which of the following experiments should he set up next (using the same cloth) to find out what he wanted to know?

A


B


C


D


Explanation:

C: Vinay wants to find out whether clothes dry faster depending on the temperature. In order to do that, he should take exactly the same cloth and put it in the same position on the clothesline on a hotter day and measure the time it takes to dry. Only option C shows the cloth placed in the same position as earlier on a hotter day.

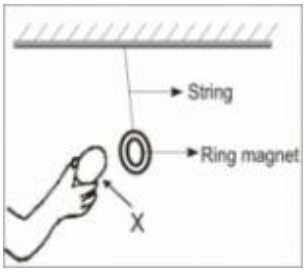
Q 5 Emperor Penguins in Antarctica are sea birds which dive under water to feed on fish, squids and krill. Which of the following MOST LIKELY helps it to catch its prey WITHOUT being caught by other predators?

A Their beaks are short and sharp like those of a sparrow.
B They have thick webbed feet which helps them to move fast over land.
C Their black and white colours makes it difficult for them to be seen in the sea.
D They have plump bodies which do not allow them to dive deep into the waters.

Explanation:

C: If a penguin wants to catch prey WITHOUT being caught by other predators, it needs to hide itself in some way. Its black and white colouring helps this happen. If the penguins are seen from above as they swim in water, they appear to match the dark colour of the sea water. If they are seen from under water, they appear to match the bright light of the sky. As a result, predators often miss seeing the penguins as they swim in water. This helps them to avoid their predators as they hunt for food.

Q 6 In the diagram shown, when an object 'X' is brought near the ring shaped magnet, the magnet moves away from it. Four friends are discussing what they can say about X.



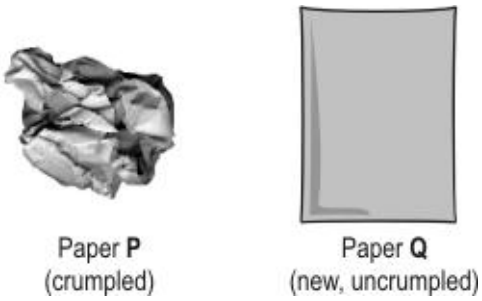
Nikhil : X can be a piece of any metal
Shailaja : X can only be a piece of iron.
Aakash : X is definitely a magnet.
Vinay : X can either be a magnet or an iron piece.
Who is correct?

A Nikhil **B** Shailaja **C** Aakash **D** Vinay

Explanation:

C: Have you played with magnets? If you have, you will know that they stick to each other and things like iron. However, sometimes, when you bring two magnets close to each other in a certain position, they move away from each other. This will never happen between a magnet and an iron piece. Try it! That's why option D is incorrect.

Q 7 Anupam takes papers P and Q, which are exactly the same. He crumples P as shown in the figure.



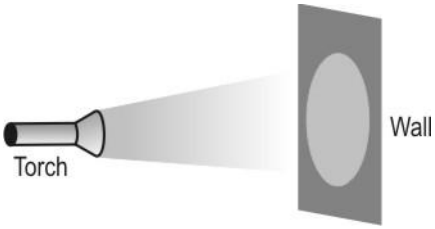
Anupam drops both P and Q from the first floor of his house. Which of the following is likely to happen?

A P will take more time to fall.
B Q will take more time to fall.
C P and Q will take equal time.
D Q will never fall as it is too light.

Explanation:

B: When paper P is crumpled, its size decreases, but the amount of paper remains the same. So if you weigh paper P and Q, they will weigh the same, but when you drop them from a height, paper Q will fall slower. This is because the amount of air pushing against paper Q will be more because it is spread out. This is the same reason why parachutes are spread out. This helps it fall at a slower pace.

Q 8 Supriya shone a torch on the wall and observed the patch of light made on the wall. Then she brought the torch closer to the wall and noted the size and brightness of the patch again. The patch of light on the wall will be_____.



A brighter and bigger
B smaller and dimmer
C brighter but smaller in size
D unchanged in size or brightness

Explanation:

C: Have you tried shining a torch beam on a wall in a dark room and observing how it looks? Try doing this. You will see that as you bring the torch closer to the wall, the patch of light will become brighter, but smaller in size. It cannot become bigger and so, option A is incorrect.

Skill: Representing, relating or explaining scientific processes or observed phenomena (44%)

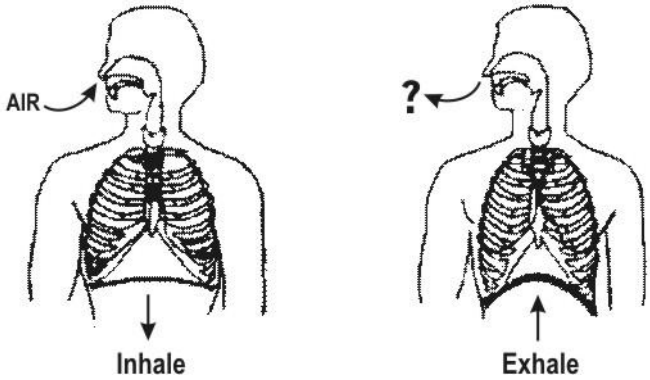
Q 1 A plant makes its own food with the help of its leaves. But from where does a seed get the food to grow into a seedling?

A It uses the stored food in the seed.
B It makes its food by photosynthesis.
C It gets it from the soil.
D It does not need food to grow.

Explanation:

A: A seed protects a baby plant inside it. The seed contains stored food to provide nourishment to the baby plant till it germinates and grows its own leaves.

Q 2 When we inhale, we breathe in air into the lungs. What do we breathe out when we exhale?

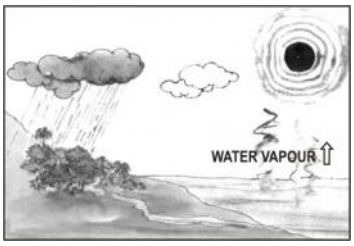


A only oxygen gas
B only carbon dioxide gas
C air that has more oxygen than inhaled air
D air that has more carbon dioxide than inhaled air

Explanation:

D: We breathe in air containing a mixture of gases. From this inhaled air only oxygen is absorbed in the lungs and utilized by the entire body. Our body gives out carbon dioxide, which is exhaled by the body along with other gases in the air. Due to this, the amount of carbon dioxide in the exhaled air is more than that in the inhaled air.

Q 3 The Water Cycle is shown here. As indicated, water from the seas and oceans evaporates and rises as water vapour:



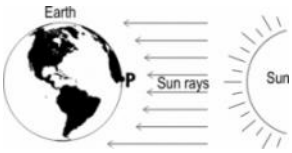
Why does water vapour rise upwards?

A It is lighter than the surrounding air.
B It is cooler than the surrounding air.
C It is attracted to the clouds in the sky.
D So that clouds can form to give us rain.

Explanation:

A: Air is made up of many different gases and is heavier compared to water vapour. As a result water vapour, on evaporating from the water bodies, rises up in the air. Clouds form when this water vapour condenses. If the water vapour had not risen up and reached a cool place, the clouds wouldn't have formed.

Q 4 After how much time will a person at point P see the Sun at the same position in the sky?



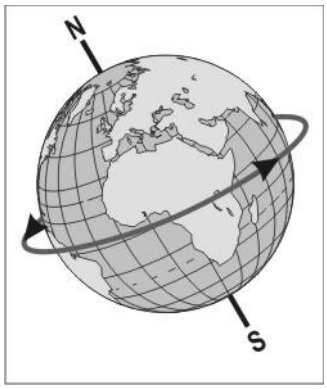
A 6 hours
B 12 hours
C 24 hours
D 30 days

Explanation:

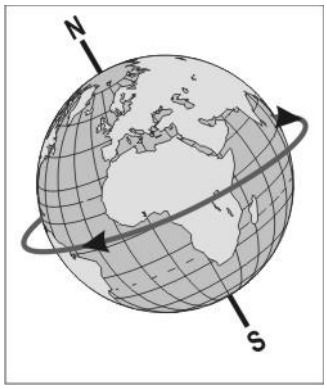
C: Point P on Earth will move as the Earth rotates on its own axis. The point will come back to the same place when one complete rotation is over. This is equal to one full day and so if a person is at point P and seeing the Sun, he will take 24 hours (1 day) to see the Sun from the same position.

Q 5 The Earth rotates on its own axis. Which of these correctly shows the direction in which it rotates?

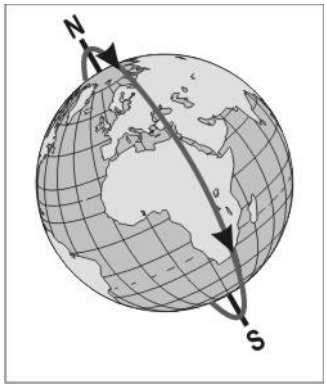
A



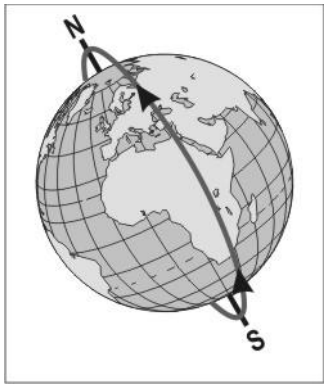
B



C



D



Explanation:

A: The Earth rotates on its own axis from West to East. You know that the Sun is fixed and the rotation of the Earth causes the Sun to appear to rise from the East and set in the West. Imagine how the Earth needs to move for that to happen.

Q 6 Plants lose excess water through parts like leaves, stems and flowers.

In order to reduce the loss of water, which of the following parts is found missing in some desert plants?

A roots **B** stem **C** seeds **D** leaves

Explanation:

D: Roots are one of the most important parts of any plant because they help in absorbing water and nutrients from the soil. So roots cannot be missing for a plant. Plants lose excess water mainly through leaves, stems and flowers and so one of these part is either missing or modified to reduce this loss of water. Mainly it is the leaves that are modified, like in case of cactus, so that they don't lose water.

Q 7 The mirror in the bathroom often clouds up while having a hot-water bath. Which of these explains this observation correctly?

A The carbon dioxide in our breath cools on touching the surface of the mirror.
B The oxygen in the atmosphere cools on touching the surface of the mirror.
C The water from the shower falls on the surface of the mirror.
D Water vapour from the shower cools on touching the surface of the mirror.

Explanation:

D: Liquid water on heating turns to gaseous water vapour. This water vapour rises up and when it comes in contact with cooler surfaces, it cools down and again turns to water. The surface of the mirror is cooler and thus, the water vapour condenses into small water droplets clouding up the mirror.

Q 8 Shyam took out a bottle of cold water from the fridge, wiped it dry and left it on the table. A few minutes later he found water droplets on the outside of the bottle. Which process leads to the formation of the droplets?

A Precipitation **B** Evaporation **C** Condensation **D** Filtration

Explanation:

C: When water vapour touches a cold surface, its state changes and it is converted to liquid water. This phenomenon is known as condensation. When the cold water bottle was kept outside, the water vapour present in the atmosphere condensed on its surface and formed the water droplets. Evaporation is when liquid water changes to gaseous form. Here, water droplets are being formed and not vapour, so B cannot be correct.

Notes

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